Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 2 · SLO 2 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 2 : SLO â€“ 2 | Stack Implementation Using Linked Lists

■ I. Push Operation â€” Insert an Element VAL

• Step 1: Allocate MEMORY for the new node and name it as NEW_NODE
• Step 2: SET NEW_NODE -> DATA = VAL
• Step 3: IF TOP = NULL
• SET NEW_NODE -> NEXT = NULL
• SET TOP = NEW_NODE
• ELSE
• SET NEW_NODE -> NEXT = TOP
• SET TOP = NEW_NODE
• [END OF IF]
• Step 4: END

■ II. Pop Operation â€” Delete an Element

• Step 1: IF TOP = NULL
• PRINT UNDERFLOW
• Goto Step 5
• [END OF IF]
• Step 2: SET PTR = TOP
• Step 3: SET TOP = TOP -> NEXT
• Step 4: FREE PTR
• Step 5: END